Differentiation and Rates of Change

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

2 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

3 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve is defined parametrically by $x = t^2 + 1$, $y = 2t^3 - 3t$. Find $\frac{dy}{dx}$ in terms of t and the value at $t = 1$.

4 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

In triangle OAB, $\overrightarrow{OA} = 3\mathbf{a}$ and $\overrightarrow{OB} = 6\mathbf{b}$. Point P lies on AB such that $AP : PB = 1 : 2$. Find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ in its simplest form.

5 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.

6 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.

7 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

A car of mass 1200 kg ascends a hill of inclination α, where $\sin\alpha = \frac{1}{20}$, against a constant resistance to motion of 500 N. Taking $g = 9.8$ m/s^2: (a) find the driving force required to travel at a steady speed of 25 m/s, (b) calculate the power developed by the engine in kilowatts.

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9 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

On an Argand diagram, the point P represents the complex number z such that $|z-(2+3i)| \leq 2$. Find the greatest possible value of $|z|$ for points in this region.

10 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Events A and B are such that $P(A)=0.6$, $P(B)=0.5$, and $P(A \cup B)=0.8$. Find $P(A \cap B)$, calculate $P(A|B)$, and determine whether A and B are independent.

11 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the equation of the perpendicular bisector of the line segment joining $A(-2,5)$ and $B(4,-3)$. Give your answer in the form $ax+by+c=0$, where a, b, c are integers.

12 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

The random variable $X \sim N(\mu, \sigma^2)$ satisfies $P(X<65)=0.8413$ and $P(X<45)=0.0228$. Find the values of μ and σ.

13 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

A solid toy is formed by attaching a cone of radius 4 cm and height 9 cm to the flat face of a solid hemisphere of radius 4 cm. Calculate the total volume of the toy. Give your answer correct to 3 significant figures.

14 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the angle θ between the vectors $\mathbf{u}=\langle 1, \sqrt{3} \rangle$ and $\mathbf{v}=\langle 2, 0 \rangle$ in degrees.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

A student calculates $\int_1^3 \frac{4}{2x+1}\,dx$ by rounding $\ln 7 \approx 1.95$ and $\ln 3 \approx 1.10$ at the start, obtaining $2(1.95-1.10)=1.70$. State the exact value of the integral and calculate the absolute error introduced by premature rounding.

16 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The masses, x grams, of 50 packets of tea gave $\sum x = 5090$ and $\sum x^2 = 518380$. (a) Calculate unbiased estimates of the population mean and variance. (b) Calculate a 95% confidence interval for the population mean.

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17 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

18 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.

19 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force F N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.

20 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

A surveyor measures distances from an observation point C to two landmarks A and B: $AC = 140$ m and $BC = 210$ m, with included angle $\angle ACB = 78^\circ$. Find the distance AB between the landmarks to the nearest tenth of a meter, and compute the area of ΔABC to the nearest square meter.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \(x \): \(y = 3x^{3} - 5x + 7 \).

a. \(\frac{dy}{dx} = 3x^{2} - 5 \)

b. \(\frac{dy}{dx} = 9x^{2} \)

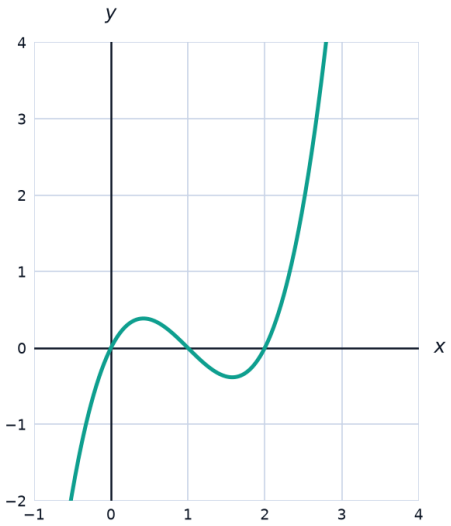
c. \(\frac{dy}{dx} = 9x^{3} - 5 \)

d. \(\frac{dy}{dx} = 9x^{2} - 5 \)

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22 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve is given by the equation $y = x^3 - 3x^2 + 2x$. Find the gradient of the curve at $x = 2$.



a. 1

b. -2

c. 2

d. 0

23 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

24 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

The curve C has equation $y = \frac{ax+6}{x^2+3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

a. $(-3, -1)$

b. $(-3, 1)$

c. $(3, 1)$

d. $(-3, -2)$

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25 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

a. 2

b. 6

c. $\frac{2}{3}$

d. $\ln 3$

26 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

a. $5e^3$

b. $6e^3$

c. $3e^3$

d. $2e^3$

27 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve satisfies $x^2 + y^2 = 25$. Find $\frac{dy}{dx}$ at the point $(3, -4)$.

a. $-\frac{3}{4}$

b. $\frac{3}{4}$

c. $\frac{4}{3}$

d. $-\frac{4}{3}$

28 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve is defined parametrically by $x = 2t^2$, $y = 3t^3$. Find $\frac{dy}{dx}$ at $t = 1$.

a. $\frac{7}{3}$

b. $\frac{9}{4}$

c. $-\frac{9}{4}$

d. $\frac{4}{9}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the gradient of the normal to $y = 3x^2 + 2x$ at $x = -1$.

a. -3/4

b. -1/4

c. 5/4

d. 1/4

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30 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the x-coordinates of the stationary points of $y = 3x^3 - 9/2x^2 - 54x$.

a. $x = -2$ or $x = -3$

b. $x = -2$ or $x = 3$

c. $x = -1$ or $x = 3$

d. $x = 2$ or $x = 3$

31 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate $y = (x)^2$ and evaluate $\frac{dy}{dx}$ at $x = 0$.

a. 4

b. 5

c. 1

d. 0

32 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the area enclosed between the curve $y = x$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 19/2

b. 9

c. 9/2

d. 7/2

33 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Given $y = 5e^{2x}$, find the coefficient of e^{2x} in $\frac{dy}{dx}$.

a. 11

b. 10

c. 2

d. 7

34 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the change in gravitational potential energy of a body of mass 18 kg raised through a height of 15 m. Take $g = 9.8$ m/s^2.

a. 5292.00

b. 2664.00

c. 2646.00

d. 2628.00

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35 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the gradient of the tangent to $y = 3x^2 - 6x$ at $x = 3$.

a. 13

b. -12

c. 11

d. 12

36 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the x-coordinate of the stationary point of $y = 3(x + 1)^2 - 2$.

a. 1

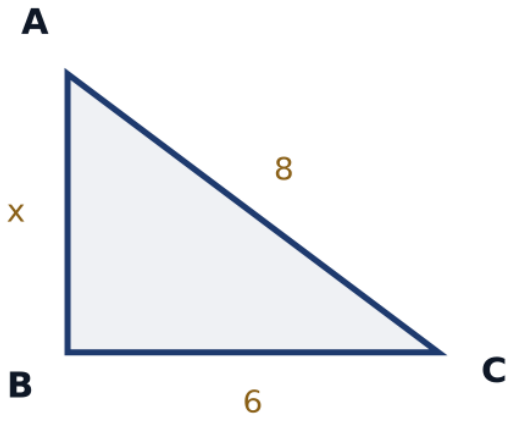
b. 0

c. -1

d. -2

37 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

In a right-angled triangle ABC, the angle B is 90 degrees. If the length of side AB is 8 units and the length of side BC is 6 units, what is the length of the hypotenuse AC?



a. 10 units

b. 14 units

c. 48 units

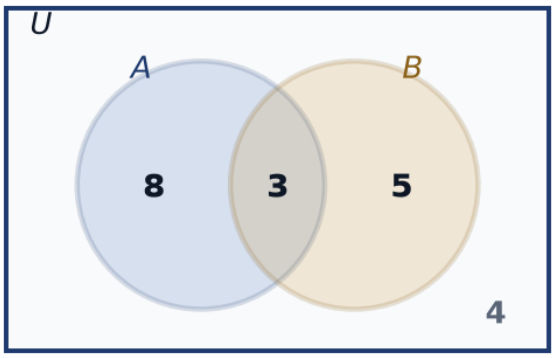
d. 20 units

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Set A has 12 elements and set B has 8 elements. If $A \cap Beq\varnothing$, what is the maximum possible number of elements in $A \cup B$?



a. 19

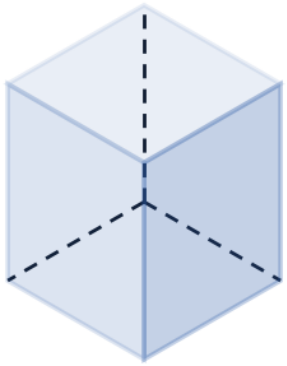
b. 20

c. 12

d. 11

39 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A rectangular prism has a length of 5 cm, a width of 3 cm, and a height of 4 cm. Calculate the volume of the prism.



a. 12 cm^3

b. 60 cm^3

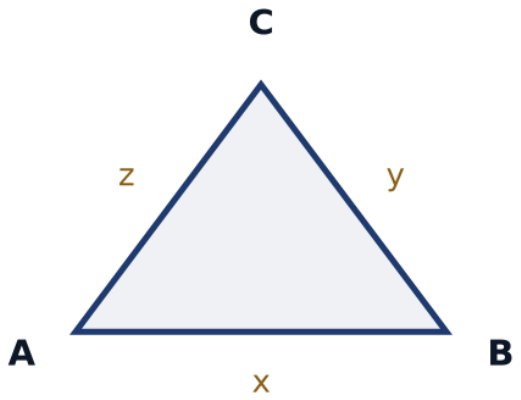
c. 47 cm^3

d. 72 cm^3

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40 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.



a. 5

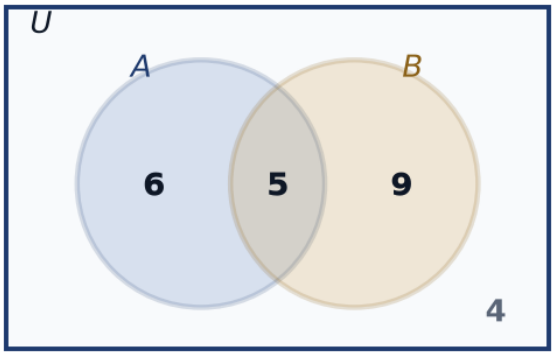
b. x

c. 6

d. 7

41 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Universal set $\mathscr{E} = \{x : x \text{ is an integer}, 1 \leq x \leq 20\}$. $P = \{x : x \text{ is a prime number}\}$ and $O = \{x : x \text{ is an odd number}\}$. What is $n(P \cap O')$?



a. 1

b. 2

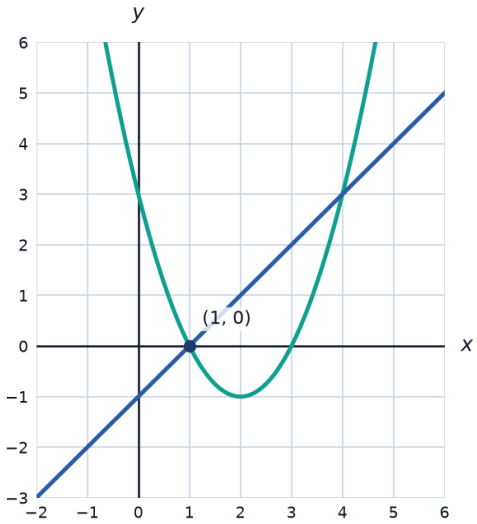
c. 8

d. 9

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42 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The diagram shows the graph of the quadratic function $y = x^2 - 4x + k$. The graph intersects the x-axis at point P(1, 0) and the y-axis at point Q. Determine the value of k.



a. k = 3

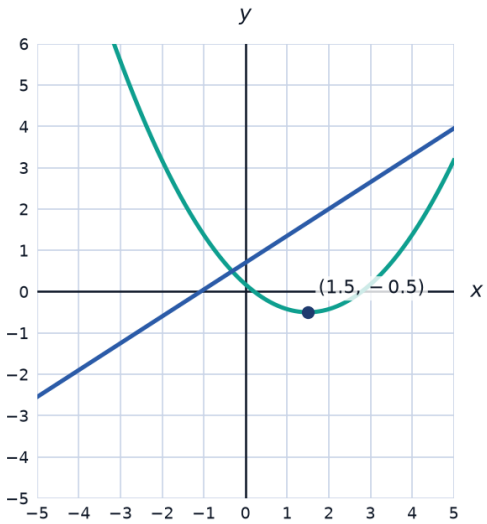
b. k = -3

c. k = 1

d. k = -1

43 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The graph of the function $y = f(x)$ is shown below. Find the value of $f(3)$.



a. 2

b. 0

c. -2

d. 4

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44 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

A number line is shown below. What is the value of the number indicated by the arrow?



a. -1

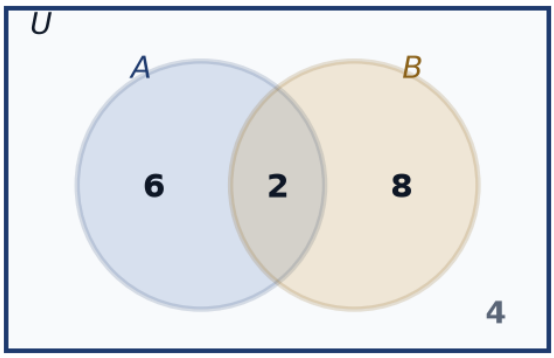
b. 0

c. 1

d. 2

45 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Universal set $\mathscr{E} = \{x : x \text{ is an integer}, 1 \leq x \leq 10\}$. Set $A = \{2, 3, 5, 7\}$. List the elements of A'.



a. $\{1, 4, 6, 8, 9, 10\}$

b. $\{4, 6, 8, 9, 10\}$

c. $\{1, 4, 6, 8, 10\}$

d. $\{2, 3, 5, 7\}$

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